

CHANG-SHENG LEE

✉ johnsonlee911205@gmail.com 🌐 johnson1205 📄 Chang-Sheng Lee

EDUCATION

National Taiwan University of Science and Technology **Sep 2021 – Jun 2025**
Bachelor of Science in Computer Science and Information Engineering **Overall GPA 4.2/4.3**

- Certificate of Achievement (Top 5% of class): Fall 2020, Spring 2021, Fall 2021, Spring 2022, Fall 2022
- Capstone Project: *"Spine2D: Skeleton Auto-Generation for 2D Animation"*
 - Awarded by National Science and Technology Council, Taiwan (Project ID: 113-2813-C-011-056-E).
 - Co-developed a plugin for automated 2D rigging by applying keypoint detection models to predict joint positions from given images.
 - Accelerated the animation production cycle from concept to final product.

RESEARCH AND WORK EXPERIENCE

Academia Sinica, Institute of Information Science **Taipei, Taiwan**
Research Assistant Jan 2025 – Present

- Researched the use of Large Language Models for detecting anomalies in firewall rule sets, and applied reinforcement learning and supervised fine-tuning to enhance model performance.
 - **Chang-Sheng Lee, Ling-Jyh Chen, "Can AI Outsmart Firewall Errors? a Study on LLMs for Anomaly Generation and Detection", 2025 IEEE Conference on Dependable and Secure Computing (DSC 2025),**

Academia Sinica, Institute of Information Science **Taipei, Taiwan**
Research Assistant Intern Jul 2023 – Jun 2024

Conducted research on infrastructure-free emergency communication systems based on LoRa, targeting disaster recovery scenarios in areas without cellular or internet connectivity. Project page: [TWC Mesh](#)

- Integrated Meshtastic, a LoRa-based mesh communication system, with ATAK (Android Team Awareness Kit) and implemented programmatic message transmission and reception using pyatak to enable collaborative offline mapping and GPS-based coordination.
- Corrected inaccurate packet format in Meshtastic documentation based on analysis using gr-lora in GNU Radio to decode LoRa signals captured via RTL-SDR and SDR-Console.
- Constructed a LoRa satellite ground station using the ESP32-based TTGO LoRa32 development board and modified a quarter-wavelength antenna to expand frequency coverage.

INDUSTRY-ACADEMIC COOPERATION PROJECTS

IGS (International Games System) – Hong Kong Mahjong AI Bot Development **Sep 2023 – Feb 2024**

- Led a 5-member bot development team to design a rule-based AI using a BK-tree data structure for efficient search of optimal winning hand combinations. (C++)

IGS – Japanese Mahjong Game Development **Oct 2022 – Aug 2023**

- Developed a scalable backend platform to manage game sessions and record game state logs. (C++)

Consumers' Foundation Chinese Taipei – Gacha Game Probability Verification **Jul 2022 – Dec 2022**

- Designed statistical models to validate gacha draw probabilities using repeated sampling and confidence intervals to determine whether actual draw rates match declared probabilities.
- Co-developed a public-facing third-party verification platform for gacha mechanics. (React, JavaScript)

IGS – Xue Liu Hong Zhong Mahjong Game Development **May 2022 – Jun 2022**

- Developed automated test tools and integrated AI bots into vendor APIs. (Python)
- Implemented Mahjong backend platform and scoring mechanisms. (C++)
- Conducted statistical validation of AI decision-making against predefined strategic rules.

EXTRACURRICULAR ACTIVITIES

ChengKong Computer Study Club, Taipei Municipal Chenggong High School **Taipei, Taiwan**
Academic Officer & Orientation Event General Coordinator Sep 2019 – Jun 2020

- Taught 80 club members C++, Data Structure, Algorithms.
- Led a collaboration across 7 high school computer science clubs to organize a joint orientation for 150 freshmen in partnership with the Ministry of Science and Technology in Taiwan, resulting in record-high club membership and increased public visibility through online media interviews.